Hao Qin

Ph.D. Candidate in Statistics, University of Arizona

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Personal Website | GitHub

Research Interests

Statistical machine learning; sequential decision-making; bandits; reinforcement learning; off-policy evaluation.

Education

University of Arizona — Aug. 2020 – Present
Ph.D. Candidate in Statistics; expected Dec. 2026 — Tucson, AZ
Advisor: Dr. Chicheng Zhang | GPA: 4.0/4.0

University of Wisconsin–Madison — Sept. 2018 – May 2020
M.S. in Statistics (GPA: 3.78/4.0) — Madison, WI

Shandong University — Sept. 2014 – Jun. 2018
B.S. in Mathematics (Grade: 91.6/100) — Jinan, China

Publications

*Equal contribution.

[1] **Optimal and Efficient Contextual Combinatorial Semi-bandits with General Function Approximation**
Hao Qin, Chicheng Zhang.
Advances in Neural Information Processing Systems (NeurIPS), 2026. Accepted. [Paper]

[2] **Taming the Monster Every Context: Complexity Measure and Unified Framework for Offline-Oracle Efficient Contextual Bandits**
Hao Qin, Chicheng Zhang.
Conference on Learning Theory (COLT), PMLR 336:5399–5464, 2026. [Paper]

[3] **Achieving Adaptivity and Optimality for Multi-armed Bandits using Exponential-Kullback Leibler Maillard Sampling**
Hao Qin, Kwang-Sung Jun, Chicheng Zhang.
Transactions on Machine Learning Research (TMLR), 2026. **Featured and J2C Certifications.** [Paper]

[4] **Physics-Informed Parametric Bandits for Beam Alignment in mmWave Communications**
Hao Qin*, Thang Duong*, Ming F. Li, Chicheng Zhang.
International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks (WiOpt), 2026. [Paper]

[5] **Harnessing Large Language Models for Virtual Reality Exploration Testing: A Case Study**
Zhenyu Qi, Haotang Li, **Hao Qin**, Kebin Peng, Sen He, Xue Qin.
Automated Software Engineering, 33, Article 7, 2026. [Paper]

[6] **Kullback-Leibler Maillard Sampling for Multi-armed Bandits with Bounded Rewards**
Hao Qin, Kwang-Sung Jun, Chicheng Zhang.
Advances in Neural Information Processing Systems (NeurIPS), 36, 2023. [Paper] [Code]

Preprints

[7] **Efficient contextual partial monitoring with general function approximation**
Kapilan Balagopalan, **Hao Qin**, Kwang-Sung Jun, Chicheng Zhang.
arXiv:2610.19804, 2026. [Paper]

Honors and Awards

NeurIPS Top Reviewer — 2024, 2026
NeurIPS Travel Grants — 2023, 2024
Carter Travel Award, University of Arizona — 2023
Third Prize, Chinese Mathematics Competition — 2016

Research Experience

Research Assistant, University of Arizona — Aug. 2020 – Present

Advisor: Dr. Chicheng Zhang | Tucson, AZ

- **Offline-oracle efficient contextual bandits.** Developed the Offline Estimation to Decisions (OE2D) framework, reducing contextual bandit learning to offline regression with near-optimal regret and $O(\log T)$ oracle calls. Introduced the Decision-Offline Estimation Coefficient to characterize the exploration–exploitation trade-off.
- **Contextual combinatorial semi-bandits.** Developed SquareCB.Comb for general reward function approximation and combinatorial action structures; established minimax-optimal regret through a computationally efficient exploration-by-optimization approach.
- **Sample-efficient multi-armed bandits.** Developed KL-MS and Exp-KL-MS for bounded rewards and one-parameter exponential families. Established asymptotic optimality, near-minimax regret, the Sub-UCB criterion, and variance-adaptive guarantees; used explicit sampling probabilities to support unbiased off-policy evaluation.
- **Physics-informed wireless learning.** Developed parametric bandit algorithms for mmWave beam alignment and tracking that exploit sparse multipath structure; evaluated them on DeepMIMO and DeepSense6G.
- **Collaborative machine learning research.** Coauthored work on LLM-based virtual reality exploration testing.

Professional Experience

Research Scientist Intern, Amazon Alexa Science — Sept. 2026 – Present

Bellevue, WA

- Build an end-to-end evaluation system for AI agents, covering metric selection, benchmark dataset development, and launch within Amazon's internal ML pipeline.

Applied Scientist Intern, Amazon Business — Jun. – Sept. 2024

Seattle, WA

- Built Amazon Business's first personalized email recommender using contextual bandits; developed the feature pipeline and recommendation policy, and evaluated performance with off-policy evaluation and an online A/B test.

Statistical Consultant, University of Arizona — Sept. 2023 – Jul. 2024

Tucson, AZ

- Advised healthcare and biology researchers on study design and hypothesis testing; analyzed high-dimensional time series and interpreted results with collaborators.

Teaching Experience

Teaching Assistant, University of Arizona — 2021 – 2025

MATH 263: Introduction to Statistics and Biostatistics; MATH 571B: Design of Experiments; CSC 352: Systems Programming and UNIX; DATA 474: Introduction to Statistical Machine Learning.

Selected Presentations

Taming the Monster Every Context	COLT 2026
Physics-Informed Parametric Bandits for Beam Alignment	WiOpt 2026
Kullback-Leibler Maillard Sampling for Multi-armed Bandits	ITA 2024
A Short Survey on Multi-Armed Bandits with Bounded Rewards	2023

Professional Service

Reviewer: NeurIPS, ICML, ICLR, AAAI, AISTATS, Transactions on Machine Learning Research, and ACM Transactions on Probabilistic Machine Learning.